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We Measure What Changes

A proportionate evidence standard for learning and skills programmes delivered by small providers in low-resource settings

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ABSTRACT

Funders, governments and employers increasingly ask training providers to show that their programmes work. In education technology the evidence base is weak and often supplied by vendors, and in workforce training most evaluation stops at participant satisfaction. This paper proposes a proportionate evidence standard for small providers in low-resource settings. It combines two established frameworks, the four levels of training evaluation and the tiered evidence definitions used in United States education law, and anchors them in the best available experimental evidence from Kenya, Sub-Saharan Africa and other low- and middle-income countries. The review finds that experimental evidence consistently favours instruction matched to the learner's level and sustained support for teachers, and that effects of technology depend on the pedagogy it carries. It also finds that small providers can meet credible standards without large budgets if they pre-specify outcomes, use comparison groups where feasible, engage an independent evaluator, disaggregate results, report costs, and publish null findings. The paper sets out an evidence ladder, a minimum reporting standard, and the threats to validity that a small provider is most likely to encounter. It closes by stating where Instracta currently sits on the ladder and what it will do to climb it.

Keywords: evidence-based education; training evaluation; impact evaluation; education technology; randomised controlled trials; Kenya; monitoring and evaluation

1. Introduction

Education technology is sold on promises of improved learning. The question that matters to a learner, a ministry or a funder is whether those promises have been tested, by whom, and against what. The 2023 Global Education Monitoring Report put the problem plainly. It noted that much of the evidence on education technology is produced by vendors, that only 7% of education technology companies in the United Kingdom had conducted randomised controlled trials, and that in a survey of 17 US states only 11% of teachers and administrators had asked for peer-reviewed evidence before adopting a product (UNESCO, 2023). It also recorded a case in which studies funded by a major publisher contested independent analysis that had found its products had no impact (UNESCO, 2023).

The same weakness exists in workforce and professional training. Programmes are commonly judged by whether participants enjoyed them. That is easy to measure and weakly related to whether anything changed. For a small provider in a low-resource setting the difficulty is compounded: rigorous evaluation appears to require resources that the provider does not have, which tempts it either to say nothing or to claim more than it can support.

This paper argues for a middle course: a proportionate evidence standard that a small provider can meet, that is honest about what each kind of evidence can and cannot show, and that is oriented towards learning outcomes rather than activity. Section 2 reviews the frameworks. Section 3 summarises what experimental evidence says about what works, with attention to Kenya. Section 4 proposes the evidence ladder. Section 5 sets out a minimum reporting standard and threats to validity. Section 6 states where Instracta stands.

2. Two frameworks: what to measure and how strong the evidence is

Two questions need separate answers: what should be measured, and how much confidence a given piece of evidence deserves. Different frameworks address each.

On what to measure, the dominant framework in training evaluation is Kirkpatrick's four levels: reaction, learning, behaviour and results (Kirkpatrick & Kirkpatrick, 2006). Reaction asks whether participants found the training engaging and relevant. Learning asks whether knowledge, skill or attitude changed. Behaviour asks whether participants apply what they learned in the workplace or classroom. Results asks whether the organisation or community outcomes the training was meant to move did in fact move. Phillips (1997) added a fifth level, return on investment. The strength of the framework is that it makes the chain of causation explicit; its weakness, widely noted, is that the levels are not a guarantee of one another, so that strong results at one level do not imply strong results at the next. A meta-analysis of organisational training by Arthur, Bennett, Edens and Bell (2003) found that training was on average effective across reaction, learning, behaviour and results criteria, and that effect sizes

varied with the training method and with which criterion was used. The practical implication is that a provider reporting only reaction data should not claim impact.

On how much confidence evidence deserves, we draw on the tiers defined in the Every Student Succeeds Act of the United States and in the guidance that implements it. Tier 1, strong evidence, requires a well-designed experimental study. Tier 2, moderate evidence, requires a well-designed quasi-experimental study. Tier 3, promising evidence, requires a well-designed correlational study with statistical controls for selection bias. Tier 4, demonstrates a rationale, requires only an intervention based on a well-specified logic model that is itself grounded in rigorous research, with a planned or ongoing evaluation (Center for the Analysis of Data in Education Research [CADRE], n.d.). The tiers were designed for US schools and are not a Kenyan or African standard. Their value here is that they are explicit, graduated, and do not demand randomisation from a provider that cannot yet afford it, while refusing to treat a logic model as proof.

3. What the experimental evidence says

A proportionate standard should be anchored in what the best studies have shown works, so that a provider's logic model does not rest on fashion. Four findings recur.

First, matching instruction to the learner's level raises learning. Banerjee, Cole, Duflo and Linden (2007) found in two randomised experiments in India that remedial teaching by community tutors raised average test scores by 0.14 standard deviations in the first year and 0.28 in the second, that a computer-assisted mathematics programme raised scores by 0.35 and 0.47 standard deviations, and that gains persisted after the programmes ended; children who experienced smaller classes alone showed no gains. Muralidharan, Singh and Ganimian (2019) found that an adaptive after-school programme raised mathematics scores by 0.36 and Hindi scores by 0.22 standard deviations over about 4.5 months, with proportionally larger gains for weaker students. In Kenya, Duflo, Dupas and Kremer (2011) found that students in schools that grouped first-grade pupils by initial achievement scored 0.14 standard deviations higher after 18 months than students in schools that did not, with the advantage persisting a year after the programme ended, and interpreted the effect as arising from teachers being able to pitch instruction at an appropriate level.

Second, teacher support matters more than materials alone. Piper, Simmons Zuilkowski and Mugenda (2014) reported on a randomised trial of the PRIMR Initiative in more than 400 Kenyan schools and found that a package of curriculum alignment, ongoing instructional support, classroom observation and books in English and Kiswahili improved oral reading fluency in both languages; they concluded that in-classroom teacher support was important. In a further analysis of the PRIMR randomised trial, Piper, Zuilkowski, Dubeck, Jepkemei and King (2018) compared combinations of teacher professional development and coaching, student textbooks, and structured teachers' guides, and found that the most expensive combination

was also the most cost-effective because its added impact outweighed its cost. Conn (2017), in a meta-analysis of 56 articles covering 66 experiments and quasi-experiments in Sub-Saharan Africa, found that programmes changing teacher pedagogy or classroom instruction had effect sizes about 0.30 standard deviations larger than all other programme types combined.

Third, teacher knowledge is a binding constraint in the region. Bold and colleagues (2017) used nationally representative surveys from seven Sub-Saharan African countries, covering close to 40% of the region's population, and found that students received about two hours and fifty minutes of teaching a day, just over half the scheduled time, that many teachers did not fully master the curriculum they taught, and that teachers' content and pedagogical knowledge had large positive effects on student achievement. For a training provider, the finding suggests that programmes aimed at teachers and trainers have a plausible route to impact, and that their evaluation should measure changes in knowledge and in practice.

Fourth, technology helps when it carries good pedagogy and is evaluated. Escueta, Nickow, Oreopoulos and Quan (2020) reviewed experimental research on technology in education and concluded that technology can help, and in some cases hinder, learning. Hennessy and colleagues (2022) produced what they described as the first systematic review of technology-mediated teacher professional development in low- and middle-income countries, noting that professional development there is often limited and short-lived and that technology may strengthen both formal programmes and informal peer learning. The review literature as a whole should be read with caution. Evans and Popova (2016) showed that six systematic reviews reached divergent conclusions mainly because they included different studies, and Glewwe and Muralidharan (2016) and Kremer, Brannen and Glennerster (2013) both stress that the evidence base remains thin for many intervention types and is uneven across contexts.

Taken together, these findings give a provider a defensible logic model: diagnose the learner's level, teach to it, support the person who teaches, and measure. They do not license the claim that any particular course works.

4. An evidence ladder for small providers

We propose that a provider position each programme on a four-rung ladder, state its rung publicly, and commit to the next rung with a date. The rungs correspond to the ESSA tiers but are described in terms of what a small provider would actually do.

Table 1. Evidence ladder and what a small provider does at each rung

Rung	ESSA tier	What the provider does	What it can claim
A. Rationale	Tier 4	Publishes a logic model linked to named research; collects reaction and learning data; plans an evaluation	The design is consistent with research. No claim of impact.
B. Before and after	Below tier 3	Pre-registers outcomes; measures learning before and after with a validated instrument; reports attrition and disaggregates by sex, disability and location	Participants improved on the measured outcome. Cause not established.
C. Comparison	Tiers 2 and 3	Adds a comparison group (waitlist or matched), checks baseline equivalence, uses statistical controls; independent analysis	Promising or moderate evidence of effect, with stated limits.
D. Randomised	Tier 1	Randomises participants or sites (a waitlist design is often feasible), pre-registers, partners with a university or research body	Strong evidence of effect in this context.

The ladder has three virtues. It makes claims proportionate to design: a provider on rung B says that participants improved, not that the programme caused the improvement. It gives a small provider a legitimate starting point, since rung A requires thought and honesty rather than money. And it is cumulative, because the instruments and data collected at rung B are the baseline for rung C.

Waitlist randomisation deserves emphasis because it is often the cheapest credible design for a training provider. When more people apply than a funded programme can serve, assigning places by lottery is a fair allocation rule as well as a research design; the lottery in Muralidharan et al. (2019) worked in exactly this way, with winners receiving a voucher and losers forming the comparison group.

5. A minimum reporting standard and the threats to

validity

Whatever the rung, we propose that a provider should meet a minimum reporting standard. The following ten items are drawn from common practice in impact evaluation and in the evidence guidance cited above.

1. State the outcome to be measured, and how, before data collection.
2. Name the instrument and say whether it was validated for the population and language.
3. Report the number enrolled, the number who completed, and the number measured at endline, so that attrition is visible.
4. Report effect sizes, not only whether a result was statistically significant, and give the uncertainty.
5. Disaggregate results by sex, disability status, age and location.
6. Report the cost per participant and, where outcomes allow, the cost per unit of learning gain.
7. Identify who conducted the analysis and whether they were independent of the delivery team.
8. Describe the comparison, or state that there was none.
9. Publish null and negative findings with the same prominence as positive ones.
10. Protect participant data in line with the Data Protection Act, 2019 of Kenya, and obtain informed consent for evaluation use.

Small providers are exposed to a particular set of threats to validity. Selection bias arises when the people who choose a programme differ from those who do not, and is the main reason before-and-after results overstate effects. Attrition biases results if those who drop out differ from those who stay, which is common in online courses with high drop-out. Novelty and attention effects can inflate early results. Measurement problems arise when the programme teaches the content of the test, when instruments are not validated, or when participants know their answers will be used to judge the programme. Finally, funder pressure can push a provider toward favourable reporting, which is why independence in the analysis, and a commitment to publish regardless of results, are included in the standard. The 2023 Global Education Monitoring Report's observation about vendor-funded studies is a warning that independence matters (UNESCO, 2023).

Independence can be achieved at a modest cost. A university research unit, a development-economics group, or a qualified independent evaluator can review the design, hold the data, and sign the analysis. The role is distinct from that of an advisory board, and its occupant should not be responsible for delivery.

6. Where Instracta stands and what it commits to

Applying the ladder honestly, and on the evidence Instracta has documented to date, its programmes sit at rung A, with some descriptive data approaching rung B. Where Instracta reports results from past work, they should be read as descriptive and not as causal. Instracta therefore commits to the following: to publish a logic model for each initiative that names the research on which it rests; to move its first sponsored cohort to rung B with pre-specified outcomes and a validated instrument; to move at least one programme to rung C using a waitlist comparison, subject to funding; to appoint an independent evaluator who does not deliver the programme; to publish cost per participant; and to publish its findings whether or not they flatter the organisation.

These commitments are modest by research standards. They are also the commitments that distinguish a provider who is trying to learn from one who is trying to sell. Their value to funders is that they make Instracta's claims checkable.

7. Limitations

This is a narrative review. It does not follow a systematic review protocol, and it draws many of its effect sizes from India, where the largest body of experimental work on technology-aided instruction exists. The Kenyan studies cited concern early-grade literacy and school-based reforms; their transfer to adult and workforce learning is plausible but unproven. The ESSA tiers originate in United States law and are used here as a convenient graduated scale, not as a local standard. The proposed ladder has not itself been validated. These limits suggest a research agenda: to develop and test a Kenyan version of the ladder with funders and evaluators, to compare reaction, learning and behaviour measures in the same programmes, and to estimate the minimum evaluation budget that a small provider needs at each rung.

8. Conclusion

Education technology will be judged, eventually, by what it changes. The evidence suggests that what changes learning is not novelty but the match between instruction and learner, and the sustained support of the person who teaches. A small provider cannot always run a randomised trial, but it can always say what it measured, how, and what it does not know. A proportionate evidence standard turns that honesty into practice: claims no larger than the design supports, a public commitment to the next rung, independent analysis, and reporting of costs and failures. That is a standard worth being held to.

Statement on method and interests

This working paper is a narrative review. Sources were identified through web and repository searches in October 2026, giving priority to international agency reports, systematic reviews and randomised or quasi-experimental studies. Bibliographic details were checked against publisher or repository records; for several sources only the abstract or catalogue record was available and the statements made here are limited to what those records report. Drafting used AI assistance (Claude, Anthropic), and the author is responsible for the content. The author is the founder of Instracta eLearning Solutions, whose programmes are discussed. The paper has not been peer reviewed.

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